FURTHER MECHANICS / CENTRE OF MASS

Centre of Mass of Systems of Particles

Mark scheme 9 questions · 59 marks

EXAM BOARDS Edexcel FM2 9FM0/4C · AQA 7367/3M · OCR A Y543 · OCR MEI Mech Minor Y431 OCR MEI Mech Major Y421 · CIE 9231 Paper 3

Questions

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	Total	**59**	

Reading the mark scheme

These rules apply throughout, unless a note says otherwise.

Marks

Every mark is worth one, so **M1 A1** on a row is two marks. Only marks coded ft follow through.

Code	Meaning
M1	Method: a correct method, applied to this question. A slip doesn't cost it if the method is clear. A formula must be used, not just quoted.
dM1	Dependent method: as M1, but only if the method mark before it was earned, unless its note names another.
A1	Accuracy: a correct answer or step. It needs the method mark it follows.
A1ft	As A1, following through from an earlier answer that the note names.
A1 AG	The last mark of a part whose answer is printed in the question: complete working, with no errors.
B1	A correct value, fact or statement. It needs no method mark.
B1ft	As B1, following through from an earlier answer that the note names.
E1	A written explanation, interpretation or conclusion. Concluding a proof by induction needs every earlier mark.

Abbreviations

ft and AG qualify a code in the Marks column. The others appear in the notes. A grey line gives the rule where the name alone doesn't.

ft	follow through Correct work from the student's own earlier wrong value earns the mark.	**AG**	answer given The answer is printed in the question, so every step to it must be shown.
cao	correct answer only No follow-through from an earlier error.	**cso**	correct solution only No error anywhere in the part, not just in its answer.
oe	or equivalent	**awrt**	answers which round to
soi	seen or implied Earned if written, or if later working shows it was used.	**isw**	ignore subsequent working A correct answer, once reached, isn't spoiled by later working.
SC	special case Credit, as the note says, for work outside the main method.	**their**	the student's own earlier value
condone	a slip that doesn't cost the mark	[ ]	working the mark doesn't need
s.f.	significant figures	**d.p.**	decimal places

Reading the scheme

- Each row is a line of working, with the marks it earns beside it. Where a row carries two marks, each note starts with its code.
- **[*n*]** ends each part with its total.
- A shaded *Alternative* is a second method for the same total. Mark an answer by the method it follows.
- Any other valid method earns the equivalent marks, unless the question names a method or says *hence*.
- A sketch earns one B1 for each feature named below its figure.

Accuracy and answers

- Unless a question says otherwise, give answers to 3 significant figures, and angles in degrees to 1 decimal place. Angles in radians take 3 significant figures.
- A final answer with more figures is accepted if it rounds to the scheme's value; one with fewer is not. An answer spoiled by rounding too early loses its accuracy mark.
- Where a question asks for an exact answer, a decimal loses the final accuracy mark. Otherwise either is accepted.
- Wrong or missing units cost nothing unless the question asks for them.
- An omitted $\mathrm{d}\theta$, $\mathrm{d}x$ or $\mathrm{d}u$ in an integral is not penalised where the variable of integration is clear from the integrand and the limits. It is penalised where it is genuinely ambiguous: a substitution part-way through, two variables in play, or limits in the wrong variable.
- A correct answer with no working earns full marks, unless the question needs the working: show that, prove, hence, a method it names, or an exact answer. A wrong answer with no working earns nothing. Calculator answers are treated the same way.
- A correct answer, once reached, isn't spoiled by later working (isw).

Value of g

- Take $g = 9.8\,\mathrm{m\,s^{-2}}$ unless a question states a value, which is then the only one accepted. Answers that use $g = 9.8$ may be given to 2 or 3 significant figures.
- Where a question doesn't state g, an answer worked with $g = 10$ throughout earns the same marks, keeping any value printed in the question. The worked solutions list the final answers with $g = 10$.

Question	Scheme	Marks	Notes
1	$20\bar{x} = 6(1) + 4(5) + 3(-2) + 5(3) + 2(0)$ $\Rightarrow \bar{x} = \ldots$	**M1**	Equates total mass times $\bar{x}$ to a sum of the nonzero $m_i x_i$ terms, the lamina at $(1, -1)$. Condone slips in values.
	$\bar{x} = \frac{7}{4}$	**A1**	oe, such as 1.75.
	$20\bar{y} = 6(-1) + 4(0) + 3(1) + 5(-4) + 2(-3)$ $\Rightarrow \bar{y} = \ldots$	**M1**	Forms the same equation for $\bar{y}$, with the same allowances.
	$\bar{y} = -\frac{29}{20}$	**A1**	oe, such as -1.45.
	Total for Question 1	**4**	

Question	Scheme	Marks	Notes
2(a)	$M\bar{x} = \sum m_i x_i$, $M = 10$ $10\bar{x} = 1(0.1) + 2(0.9) + 3(0.7) + 4(0.2)$	**M1**	Forms $M\bar{x} = \sum m_i x_i$ with all four terms and the total mass. Condone slips in values.
	$\bar{x} = 0.48$	**A1**	
	$10\bar{y} = 1(0.1) + 2(0.1) + 3(0.5) + 4(0.6)$	**M1**	Forms the same equation for $\bar{y}$. Condone slips in values.
	$10\bar{y} = 4.2 \Rightarrow \bar{y} = 0.42$	**A1 AG**	Correct substituted equation leading without error to $\bar{y} = 0.42$; combined arithmetic accepted.
		[4]	
2(b)	Peg must be at G, vertically below the centre of mass, but $G(0.48, 0.42)$ lies inside the quadrilateral, not on any rod	**E1**	States that the peg must be at G and that G is not on the frame. Both needed: the first may be implied by the conclusion, and G inside the quadrilateral suffices for the second.
		[1]	
	Total for Question 2	**5**	

Question	Scheme	Marks	Notes
3	$10m\bar{x} = 2m(-1) + 4m(1) + 3m(4) + m(6)$	**M1**	Equates total mass times $\bar{x}$ to a sum of four mass-times-x terms. Condone slips in values; m may be omitted throughout.
	$\bar{x} = 2$	**A1**	
	$10m\bar{y} = 2m(3) + 4m(-1) + 3m(2) + m(-2)$	**M1**	Equates total mass times $\bar{y}$ to a sum of four mass-times-y terms. Condone slips in values; m may be omitted throughout.
	$\bar{y} = \frac{3}{5}$	**A1**	oe, such as 0.6.
	Total for Question 3	**4**	

Question	Scheme	Marks	Notes
4(a)	$M\bar{x} = \sum m_i x_i$: $(7+\lambda)m \times 2 = 2m(2) + 4m(1) + \lambda m(4)$	**M1**	Forms a moments equation for x with total mass $(7+\lambda)m$, $\bar{x} = 2$ and all three nonzero terms. Condone slips in values; m may be omitted throughout.
	$14 + 2\lambda = 8 + 4\lambda \Rightarrow \lambda = 3$	**A1 AG**	Correct equation leading without error to $\lambda = 3$; combined steps accepted.
		[2]	
4(b)	$10mk = 2m(3) + 4m(-1) + 3m(2)$	**M1**	Forms a moments equation for y with total mass $10m$, $\bar{y} = k$ and all three nonzero terms. Condone slips in values; m may be omitted throughout.
	$k = \frac{4}{5}$	**A1**	oe, such as 0.8.
		[2]	
	Total for Question 4	**4**	

Question	Scheme	Marks	Notes
5	$(12+m)\bar{x} = 3(1) + 4(4) + 5(10) + 2m$	**M1**	Equates total mass, including m, times $\bar{x}$ to the sum of the four moments, including $2m$. Condone slips in values.
	$(12+m)\bar{y} = 3(9) + 4(2) + 5(7) + 5m$	**M1**	Forms the $\bar{y}$ equation as for $\bar{x}$, with the y-coordinates, including $5m$.
	$\bar{x} = \frac{69+2m}{12+m}$, $\bar{y} = \frac{70+5m}{12+m}$	**A1**	Both correct, oe; need not be simplified.
	$\text{“}\frac{70+5m}{12+m}\text{”} = 2\left(\text{“}\frac{69+2m}{12+m}\text{”}\right) - 3$ $\Rightarrow m = \ldots$	**dM1**	Substitutes their $\bar{x}$ and $\bar{y}$, each linear in m over their total mass, into $y = 2x - 3$ and solves for m. Condone sign slips.
	$m = 8$	**A1**	
	$\left(\frac{17}{4}, \frac{11}{2}\right)$	**A1**	oe, such as $(4.25, 5.5)$.
	Total for Question 5	**6**	

Question	Scheme	Marks	Notes
6(a)	$M\bar{x} = \sum m_i x_i$, where $M = \sum m_i = (6+k)m$ $(6+k)m(2a) = -am + 4am + k\mu am$	**M1**	Forms an x-moments equation with total mass $(6+k)m$, $\bar{x} = 2a$ and all three nonzero terms. Condone slips in values; m and a may be omitted throughout.
	$k\mu = 9 + 2k$	**A1**	Correct equation in k and μ, oe.
	$(6+k)m(3a) = am - 4am + 2k\mu am$	**M1**	Forms a y-moments equation with total mass $(6+k)m$, $\bar{y} = 3a$ and all three nonzero terms. Condone slips in values; m and a may be omitted throughout.
	$2k\mu = 21 + 3k$	**A1**	Correct equation in k and μ, oe.
	$2(\text{“}9 + 2k\text{”}) = \text{“}21 + 3k\text{”} \Rightarrow k = \ldots$	**dM1**	Eliminates $k\mu$ between their two equations, each linear in k and $k\mu$, to reach a value of k.
	$k = 3$	**A1**	cso.
		[6]	
6(b)	$3\mu = 9 + 2(3)$	**M1 A1**	**M1** Substitutes their k from (a) into either moments equation, or a rearrangement of it, to form an equation in μ. **A1** Correct equation in μ, oe such as $6\mu = 21 + 9$.
	$\mu = 5$	**A1**	
		[3]	
	Total for Question 6	**9**	

Question	Scheme	Marks	Notes
7(a)	$M\bar{x} = \sum m_i x_i$, $M = (k+3)m$ $(k+3)m\bar{x} = m(a) + 2m(-2a) + km(2a)$	**M1**	Uses $M\bar{x} = \sum m_i x_i$ with $M = (k+3)m$ and all three terms. Condone slips in values; m and a may be omitted throughout.
	$\bar{x} = \dfrac{a(2k-3)}{k+3}$	**A1**	oe, free of m.
	$(k+3)m\bar{y} = m(0) + 2m(3a) + km(-a)$	**M1**	Uses $M\bar{y} = \sum m_i y_i$ with $M = (k+3)m$ and both nonzero terms. Condone slips in values; m and a may be omitted throughout.
	$\bar{y} = \dfrac{a(6-k)}{k+3}$	**A1**	oe, free of m.
		[4]	
7(b)	$\left(\text{“}\frac{a(2k-3)}{k+3}\text{”} - a\right)^2 + \left(\text{“}\frac{a(6-k)}{k+3}\text{”}\right)^2$ $= \frac{1}{2}a^2$	**M1**	Substitutes their $\bar{x}$ and $\bar{y}$, in terms of a and k, into a circle equation of the form $(x-a)^2 + y^2 = r^2$.
	$4(k-6)^2 = (k+3)^2$	**A1**	Any correct equation in k alone, such as $3k^2 - 54k + 135 = 0$.
	$k^2 - 18k + 45 = 0 \Rightarrow k = \ldots$	**dM1**	Rearranges to a three-term quadratic in k and solves it (factors giving the constant term, the formula, completing the square or a calculator), or square-roots both sides; $\pm$ not needed for this mark.
	$k = 3$, $k = 15$	**A1**	Both values and no others.
		[4]	
	Total for Question 7	**8**	

Question	Scheme	Marks	Notes
8	$10m\bar{x} = 2m(1) + 5m(-3) + 3mq$ $10m\bar{y} = 2m(-4) + 5m(2) + 3m(2q-1)$	**M1**	Forms a moments equation for at least one coordinate, with all three terms and total mass $10m$. Condone slips in values; m may be omitted throughout.
	$\bar{x} = \frac{3q-13}{10}$	**A1**	oe
	$\bar{y} = \frac{6q-1}{10}$	**A1**	oe
	$d^2 = \left(\text{“}\frac{3q-13}{10}\text{”} - 1\right)^2 + \left(\text{“}\frac{6q-1}{10}\text{”}\right)^2$	**M1**	Uses Pythagoras for the distance, or its square, from $(1,0)$ to their $(\bar{x},\bar{y})$, each linear in q, or makes that join perpendicular to their line of centres. Condone sign slips.
	$100d^2 = 45q^2 - 150q + 530$	**A1**	Any correct multiple, such as $d^2 = 0.45q^2 - 1.5q + 5.3$, or $15q - 25 = 0$ by perpendicularity.
	$45\left(q - \frac{5}{3}\right)^2 + 405$ or $90q - 150 = 0$ $\Rightarrow q = \ldots$	**dM1**	Minimises their quadratic by completing the square, setting a derivative of d or d^2 to zero, or $q = -\frac{b}{2a}$, or solves their perpendicularity equation. Not for solving quadratic $= 0$.
	$q = \frac{5}{3}$	**A1**	oe, or awrt 1.67.
	Total for Question 8	**7**	

Question	Scheme	Marks	Notes
9(a)	$p - q - 3 = 0$, $p + 2q - 6 = 0$ $\Rightarrow p = \ldots$, $q = \ldots$	**M1**	Sets the resultant component to zero in both directions, with all three forces in each. Condone sign slips.
	$p = 4$, $q = 1$	**A1**	
	Moments about O, anticlockwise positive: $2q(1) + q(1) + 3(1) - 6t = 3$	**M1 A1**	**M1** Takes moments of all three forces about any point and equates them to the couple. Condone sign slips. **A1** Correct equation about their point, with or without $q = 1$ substituted.
	$t = \frac{1}{2}$	**A1**	
		[5]	
9(b)	$4\bar{x} = 2(2\lambda^2) + 1(2) + 1(0)$	**M1**	Uses $M\bar{x} = \sum m_i x_i$ with the three particles only and $M = 4$. Condone slips in values.
	$\bar{x} = \lambda^2 + \frac{1}{2}$	**A1**	oe
	$4\bar{y} = 2(\lambda) + 1(0) + 1(1)$	**M1**	Uses $M\bar{y} = \sum m_i y_i$ with the three particles only and $M = 4$. Condone slips in values.
	$\bar{y} = \frac{2\lambda+1}{4}$	**A1**	oe, such as $\frac{\lambda}{2} + \frac{1}{4}$
		[4]	
9(c)	$\lambda = \frac{4\bar{y}-1}{2}$ $\Rightarrow \bar{x} = \left(\text{“}\frac{4\bar{y}-1}{2}\text{”}\right)^2 + \frac{1}{2}$	**M1**	Eliminates λ between their $\bar{x}$, quadratic in λ, and their $\bar{y}$, linear in λ, or substitutes both into $x = 4y^2 - 2y + k$.
	$\bar{x} = 4\bar{y}^2 - 2\bar{y} + \frac{1}{4} + \frac{1}{2}$	**A1**	Reaches $4y^2 - 2y$ plus a constant from correct coordinates, or shows the λ terms cancel, without error. Condone $\bar{x}$, $\bar{y}$ for x, y.
	$k = \frac{3}{4}$	**A1**	oe, cso
		[3]	
	Total for Question 9	**12**	